ELEC5506: Process instrumentation & Control
LAB 3: Simple working elevator system

Assessment: Individual lab reports should be uploaded in LMS two weeks after the lab. The reports should include the program, a short description of rungs and observation of any possible program weakness and improvement suggestions.

YOU WILL BE ASSESSED DURING THE LAB ITSELF AS WELL, MAKE SURE YOU COME PREPARED.

1. Lab objective and scope:

Objective:

You will be returning to the elevator system from Lab 1. In this lab, you will extend your skills by creating a program for a simple working elevator.

Scope:

The program must be structured as main body and subroutines and be able to:

1) Accept request from any floor and move the car from the current level to requested level.
2) Light on the UP or DOWN arrow depending on the direction of movement.
3) The car should move with “half-speed” between levels and reduce the speed to “quarter speed” when gets close to the requested level.
4) After arriving to the requested level, the ring should bell once and the door open and stay opened for 4 seconds.

The details of the program requirements are in the following section.

2. Program structure and detailed requirements:

The final goal of this lab is to have a working elevator system that accepts only 1 request at a time. This means, when the system is executing an instruction, any new requests will be ignored until the task is completed.

The requests are sent to the system using any floor call buttons or elevator car buttons to move the elevator to the requested floor. Once the elevator reaches the floor, the system will ring the bell, open the door for 4 seconds, close the door and wait for the next instruction. A basic state diagram for the program in show in Figure 1.

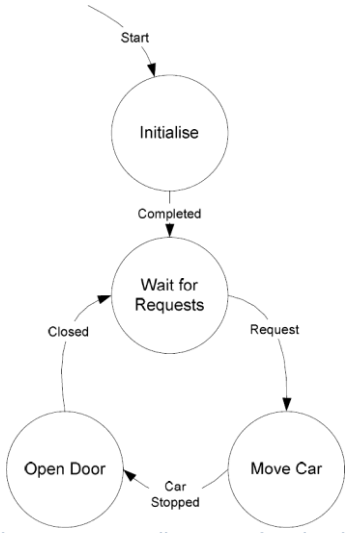

Figure 1: State diagram of a simple working elevator

You will use Jump to subroutines (JSR) instruction for each of the states. The exercises are split into each one of these states so that you approach your final goal in a structured manner. Hence, it is recommended that you test each one of the programs individually and make sure they work.

Always keep in mind that you will combine the programs at the end, so plan from the start how all the programs will work together (e.g.: using the right bit addresses for the flow of the program).

Exercise 1: (5%)
Create a program for "Initialise". This program is to be used whenever the system is restarted to ensure the elevator is in a known state. This program will simply ensure the door is closed, set the direction to down and move at quarter speed until it detects the floor 1 sensor.
- It does **not** need to stop flush with floor 1, just until a floor 1 sensor is detected.
- The door **Must** close before the elevator moves

Exercise 2: (20%)
Create a program for "Accept Request". This program should be able to get request from all four levels and determine which floor it is originated from (destination) and turn on the destination indicators for the requested level.

For example, assuming the car is at Level 1, pressing the "Level 2 car button" OR "Level 2 floor request UP" OR "Level 2 floor request DOWN" should:

- Memorize floor 2 is requested
- Light Floor 2 UP_DOWN Lamp
- Light car button level 2 Lamp

You can use Integers (INT) to indicate the requested level and compare it with the current level in the next exercise. You are free to try any other methods as well.

Exercise 3: (40%)
Create a routine for “Move Car”.
This program should:

- Determine the car direction to move from current location to the requested floor (destination).
- Light “UP” or “DOWN” arrow depending on direction of movement
- Start Moving at “Half Speed” and then reduce to “Quarter Speed” when the floor sensor detects the first finger of the gate, and then STOP at the center finger.

Exercise 4: (20%)
Create a routine for “Door Sequence”.
When the elevator has reached the intended destination:

- The bell should ring once
- The door should stay opened for 4 seconds, and then closed
- This sequence should also start if the request comes from the same level as the elevator. For example, if the request came from Level 1 floor button when the elevator is already at Level 1.

Exercise 5: (15%)
Implement a new program that combines all the programs that you have created before as subroutines into one main program. You will need to modify them so that they are only called at the start of the program and do not interfere with normal operation (it is easiest to design with this in mind from the start).

NOTE: YOU MUST USE SUBROUTINES, NO MARK WILL BE AWARDED OTHERWISE

General overview of the working elevator system

- There are 4 levels the elevator car can be moved around
- All the floor call buttons, and elevator car buttons can be used to request the elevator.
- To move the elevator, you have to latch the brake and the motor.
- There are “UP” and “DOWN” LEDs to indicate the moving direction of the elevator
- A bell is available to announce the arrival of the elevator at a floor.
- A description of the floor sensing system and a diagram of the entire elevator system are given on the following page.

Floor sensor system:
It should be considered that, in this model, in order to reduce the number of PLC inputs, a single sensor for each floor is combined with the three-finger gate mounted to the car and it is important to understand how a floor level sensor operates. **Error! Reference source not found.** shows the design of the car floor level sensor. The car movement gate is fixed to the car and a light sensor is fixed at each of the four floor levels. As the car reaches a floor the car movement gate interrupts the light beam to the light sensor. A signal as shown on the right hand side of the of the diagram is presented to a PLC input, changing state as the light beam to the sensor is interrupted by the car passing a floor level. In the real world, separate sensors would be used for above, below and at each floor.

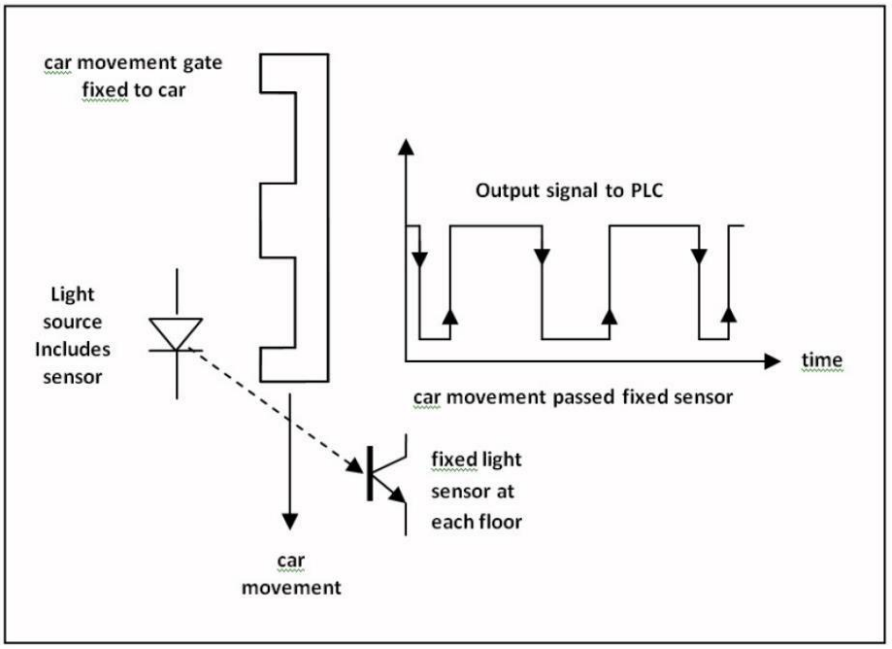

Figure 2: "three-finger gate" system

Elevator diagram:

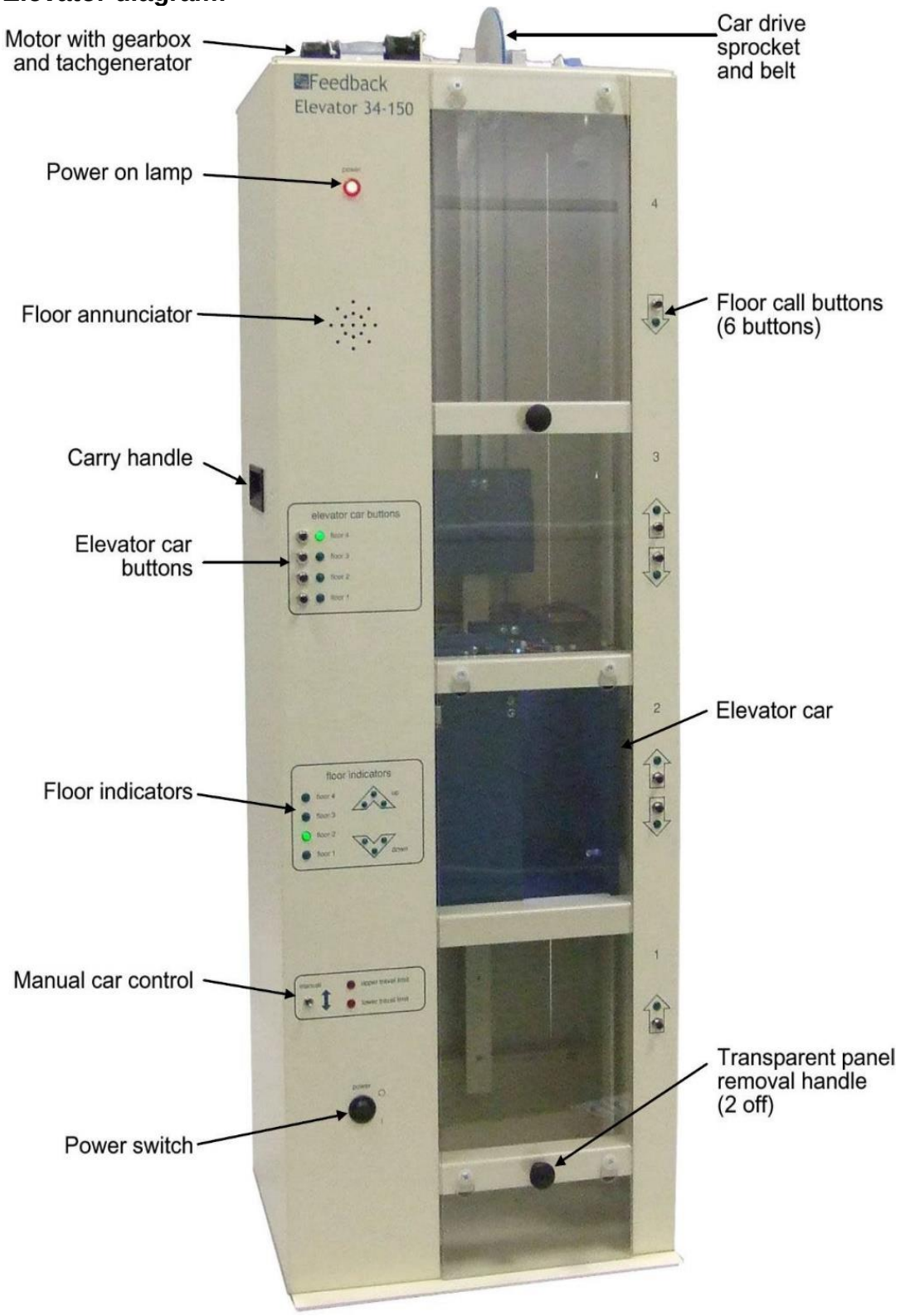

Figure 3: Entire elevator system